

A geometric interpretation of Vieta jumping

IMO 1988 Problem 6

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September 2026

The jump is a leftover width. Placing k copies of an $a \times b$ rectangle alongside one another and removing a b -square produces the width $ka - b$. The construction gives an area interpretation of the classical Vieta descent. This note makes the integer and termination arguments explicit; no claim of priority is made.

1 The problem and the strip

Problem. Let a, b be positive integers and suppose that $k = (a^2 + b^2)/(ab + 1)$ is an integer. Prove that k is a perfect square [1].

Order the lengths so that $1 \leq a \leq b$. Then $k \geq 1$ and

$$a^2 + b^2 = kab + k.$$

The two squares have the same total area as k copies of an $a \times b$ rectangle and k unit squares. The rectangles, each of width a and height b , form a strip of width ka . Define $c = ka - b$.

Why the square fits. We must first show $c \geq 0$. If $c < 0$, integrality gives $b - ka \geq 1$, so

$$k = a^2 + b(b - ka) \geq a^2 + b > b.$$

But $b \geq ka + 1 \geq k + 1$, a contradiction. Thus c is a nonnegative integer.

A new side appears in the leftover strip

Example: $a = 8, b = 30, k = 4$. Dashed lines mark the original rectangles.

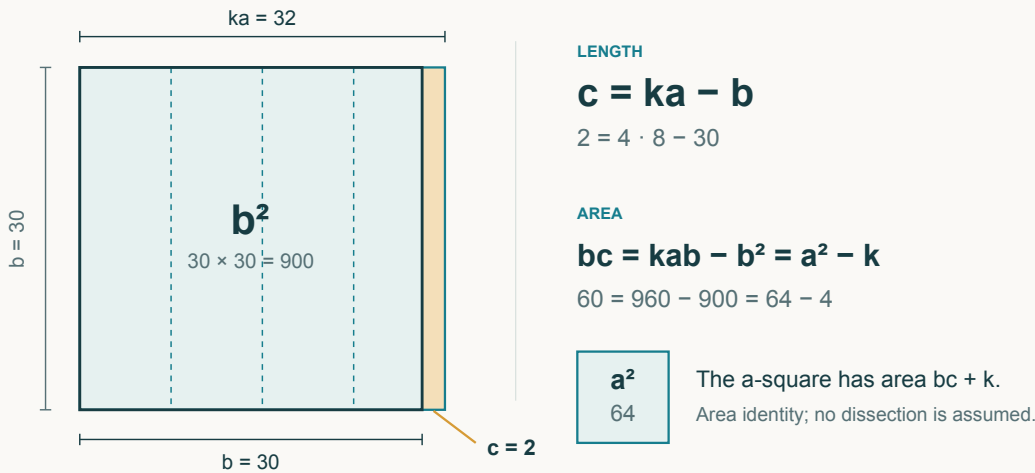


Figure 1. The leftover width is the Vieta jump. The displayed example has $a = 8, b = 30, k = 4$.

Removing the $b \times b$ square leaves a $b \times c$ rectangle. Its area is

$$bc = kab - b^2 = a^2 - k, \quad \boxed{a^2 = bc + k.}$$

This is an equality of areas. It does not assert that the long leftover rectangle fits inside the small square without being cut.

2 The next smaller pair

The strip gives $b + c = ka$ and $bc = a^2 - k$. Consequently

$$\begin{aligned} a^2 + c^2 &= bc + k + c^2 \\ &= c(b + c) + k \\ &= kac + k = k(ac + 1). \end{aligned}$$

Hence the pair (c, a) satisfies the same equation, with the same k . In area terms, the new squares match k new $c \times a$ rectangles together with the same k units of area.

Moreover,

$$0 \leq c = \frac{a^2 - k}{b} < \frac{a^2}{b} \leq a.$$

Thus the move

$$(a, b) \mapsto (ka - b, a) = (c, a)$$

strictly decreases the sum of the lengths. When $c > 0$, the new pair is ordered, positive, and integral, so the same construction applies again.

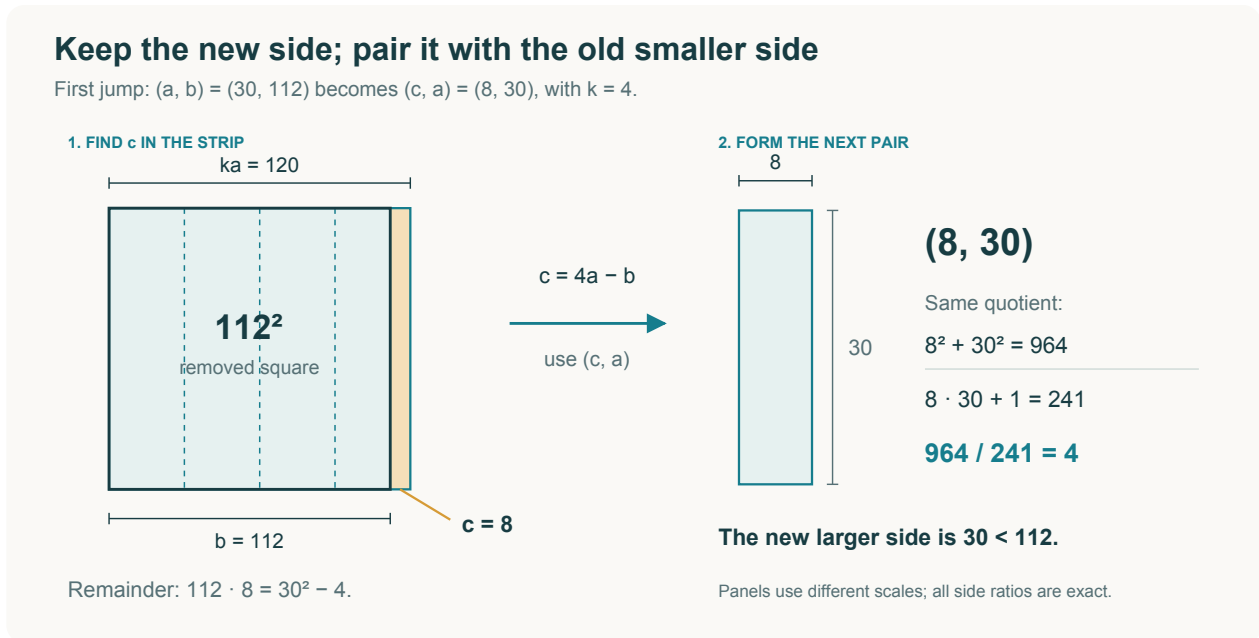


Figure 2. A first descent with $k = 4$: $(30, 112) \mapsto (8, 30)$.

3 Why the descent terminates

There cannot be an infinite strictly decreasing sequence of positive integer sums. Every pair with two positive lengths admits the next move, so eventually one length becomes zero. Write the final pair as $(0, x)$, where $x > 0$ is an integer. The preserved equation is then

$$0^2 + x^2 = k(0 \cdot x + 1), \quad \boxed{k = x^2}.$$

At the final construction the strip and the large square coincide. The leftover rectangle has zero width, and the other square has area k . This proves the claim. The case $k = 1$ is included: $(1, 1) \mapsto (0, 1)$. $\square$

A stronger conclusion. Since $\gcd(ka - b, a) = \gcd(a, b)$, each move preserves the greatest common divisor. At the endpoint $\gcd(0, x) = x$. Therefore $k = \gcd(a, b)^2$, a strengthening also given in Campbell's 1988 treatment [2].

4 Four rectangles at every step

For $(a, b) = (30, 112)$, the quotient is $13444/3361 = 4$. The complete descent is

$$(30, 112) \mapsto (8, 30) \mapsto (2, 8) \mapsto (0, 2).$$

The area identities are $900 = 896 + 4$, then $64 = 60 + 4$, then $4 = 0 + 4$. The integer k stays fixed while the lengths shrink.

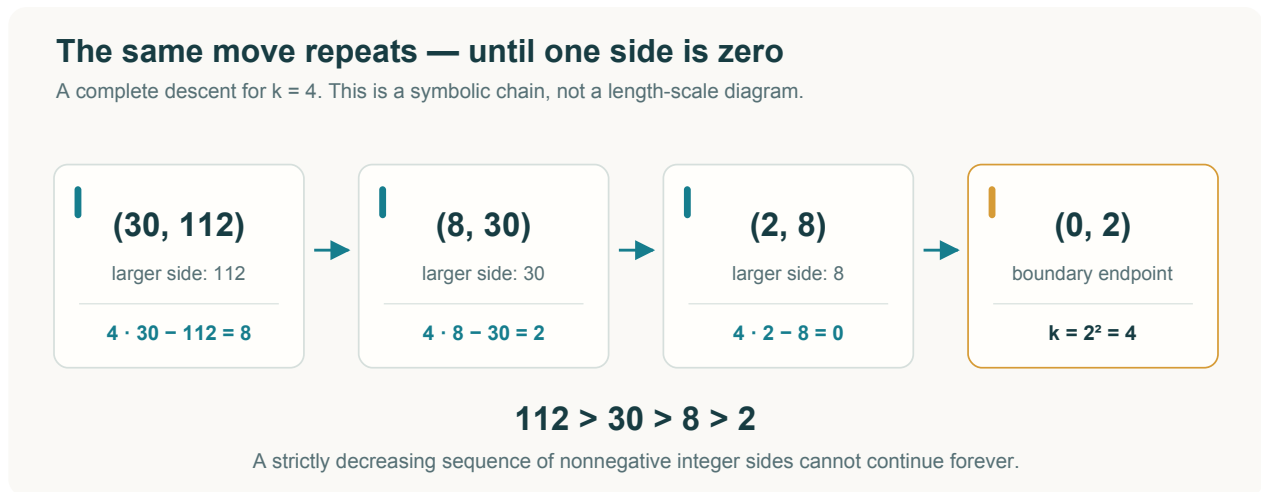


Figure 3. The complete $k = 4$ descent, shown as a symbolic chain.

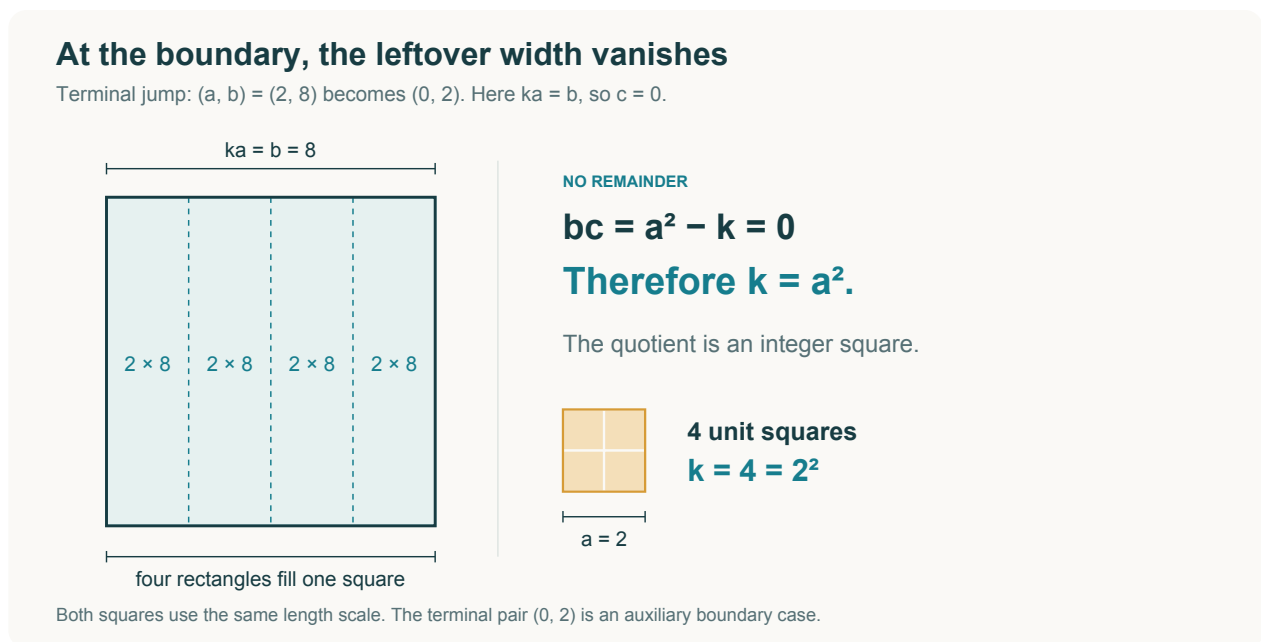


Figure 4. The final jump: $(2, 8) \mapsto (0, 2)$. The remaining square has area $4 = 2^2$.

Reading the diagrams. Each strip is drawn in proportion to its labelled dimensions. Separate constructions may have different scales to keep their geometry legible. The four unit squares at the endpoint are literal unit cells; earlier area comparisons do not specify a dissection of the smaller square.

5 Relation to the standard proof

Regarding the original equation as a quadratic in its second coordinate gives

$$t^2 - kat + (a^2 - k) = 0.$$

The roots are b and $ka - b$. Exchanging them is precisely Vieta jumping. The strip construction makes the second root visible as a remaining width. The number-theoretic mechanism is the classical descent; this note develops its repeated-rectangle interpretation.

6 Context and references

The arithmetic descent is established prior work. Campbell's early treatment includes the greatest-common-divisor conclusion [2]. Wu presents a geometric visualization using integer points on a hyperbola, including the $k = 4$ numerical orbit used here [3]. A 2016 online discussion examines a remainder-style descent and includes a separate geometric attempt [4]; that discussion is not a premise of the proof given in this note.

The rectangle strip provides an area interpretation of the same transformation. These references establish context, not a comprehensive priority investigation. No claim is made that this particular presentation is historically new.

- [1] International Mathematical Olympiad. *1988 problems*, English, Problem 6.
[Official problem paper](#).
- [2] John Campbell. "A Solution to 1988 IMO Question 6." *Mathematics Competitions* **1**(2), 1988, pp. 29–32.
[Original article](#).
- [3] Kyle Wu. "Vieta jumping: visualization and intuition." *Parabola* **60**(1), 2024.
[Publisher's PDF](#).
- [4] Rutger Moody and contributors. "Math Olympiad 1988 problem 6, canonical solution 2 without Vieta jumping." *Mathematics Stack Exchange*, 2016.
[Discussion of related approaches](#).

About this note. The repeated-rectangle interpretation was suggested by Pranay Manocha. The repository's README contains the complete visual exposition; this note is its typeset companion. SVG diagrams and their vector PDF exports share the same artwork. LaTeX source, diagram-generation scripts, and finite arithmetic checks accompany the note.

Repository: github.com/pranman/imo-1988-6-geometric-proof

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